

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 14

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

BLANK PAGE

1 By completing the square, find the minimum value of $2x^2 - 12x + 23$.

A -13

B 5

C 23

D 7

E -5

F 3

2 The perpendicular bisector of the line segment joining (1, 5) and (7, 1) has equation

A $3x - 2y = 6$

B $2x - 3y = -1$

C $3x + 2y = 18$

D $2x + 3y = 17$

E $y = \frac{3}{2}x - 3$

F $3x - 2y = 18$

3 Find the value of $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$.

- A $\frac{9}{4}$
- B $-\frac{4}{9}$
- C $\frac{4}{9}$
- D $-\frac{9}{4}$
- E $\frac{2}{3}$
- F $\frac{3}{2}$

4 $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$. Which of the following statements is/are true?

- I** A and B are independent.
- II** A and B are mutually exclusive.
- III** $P(A \text{ or } B) = 0.8$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 5 Claim: if x and y are real with $x^2 = y^2$, then $x = y$.

Which of the following is a counterexample to the claim?

- A** $x = -1, y = -1$
- B** $x = 1, y = 2$
- C** $x = 2, y = -2$
- D** $x = 0, y = 0$
- E** $x = 2, y = 2$

6 The statement '**if a triangle is equilateral then it is isosceles**' is true. Which of the following must therefore also be true?

- I** If a triangle is isosceles then it is equilateral.
- II** If a triangle is not isosceles then it is not equilateral.
- III** If a triangle is not equilateral then it is not isosceles.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 7 Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 8** Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?
- A** 51
- B** 2
- C** 100
- D** 99
- E** 52
- F** 50

9 For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 10** T_n denotes the n th triangular number $\frac{n(n+1)}{2}$. A student observes $T_1 + T_2 = 4$, $T_2 + T_3 = 9$, $T_3 + T_4 = 16$ and conjectures that $T_n + T_{n+1}$ is always a perfect square. Which of the following proves the conjecture?

A $T_n + T_{n+1} = \frac{n(n+1) + (n+1)(n+2)}{2} = \frac{(n+1)(2n+2)}{2} = (n+1)^2$

B $T_n + T_{n+1} = 2T_n + (n+1)$, which is a square.

C Triangular numbers are always squares.

D $T_4 + T_5 = 25$, which is a square, so the pattern continues.

E $T_{n+1} - T_n = n+1$, so the sum is a square.

11 Which of the following claims is FALSE, and therefore can be disproved by a counterexample?

- A** If $a > b$ then $a + c > b + c$, for all real a, b, c .
- B** If $a > b$ then $a^2 > b^2$, for all real a, b .
- C** If $a > b > 0$ then $a^2 > b^2$.
- D** If $a > b$ then $a^3 > b^3$, for all real a, b .
- E** If $a > b$ then $-a < -b$, for all real a, b .

12 Which of the following statements is/are true for all positive real x and y ?

I $\log_{10} (x + y) = \log_{10} x + \log_{10} y$

II $\log_{10} (xy)^2 = 2\log_{10} x + 2\log_{10} y$

III $\log_{10} x - \log_{10} y = \log_{10} x \div \log_{10} y$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

13 A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I** The angle opposite c is obtuse.
- II** c is the longest side.
- III** The triangle cannot be isosceles.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 14** The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

I $a \leq 3$

II $c \geq 5$

III $b = 4$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

15 The function f has $f'(x) = (x - 1)(x - 3)^2$. Which of the following statements is/are true?

- I** f has a local minimum at $x = 1$.
- II** f has a local maximum at $x = 3$.
- III** f is increasing for all $x > 1$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 16** A statement reads: ****for every positive integer n , there is a prime p with $n < p < 2n$.**** Which of the following is its negation?
- A** For every positive integer n , there is a prime p with $p \leq n$ or $p \geq 2n$.
 - B** There is a positive integer n such that no prime p satisfies $n < p < 2n$.
 - C** There is a positive integer n and a prime p with $p \leq n$ or $p \geq 2n$.
 - D** There is a prime p such that for every positive integer n , $n < p < 2n$ fails.
 - E** For every positive integer n , no prime p satisfies $n < p < 2n$.

17 Which of the following statements is/are true for every function f that has a derivative at every real number?

I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.

II If f has a local minimum at $x = a$ then $f'(a) = 0$.

III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

18 Which of the following statements can be disproved by a single counterexample?

I Every prime number greater than 2 is odd.

II Every integer of the form $4k + 3$ is prime.

III There exists an even prime number.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

19 Given that $|2x - 3| < 5$, which of the following statements is/are true?

I $-1 < x < 4$

II $x^2 < 16$

III $x > -1$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

20 The integers a, b, c satisfy $a + b + c = 0$ and $abc = 12$. Which of the following statements must be true?

I Exactly one of a, b, c is positive.

II $a^2 + b^2 + c^2$ is even.

III One of a, b, c is divisible by 3.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST